

3. [Maximum mark: 4]

The random variable X is normally distributed with mean 10 and standard deviation 2.

- (a) Find the probability that X is more than 1.5 standard deviations above the mean. [2]

The probability that X is more than k standard deviations above the mean is 0.1, where $k \in \mathbb{R}$.

- (b) Find the value of k . [2]

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